Notes for students about Laboratories

Lab Report

- Understand the task before starting!
- 5-10 pages
- Give feedback on errors and difficulties!
- Scientific spelling/style:
  - $T = 10 - 60\ °C$
  - Physical quantities and variables in italics
  - Space before units (in LaTeX ~ or \,)
  - **Units in upright font (to distinguish them from variables)**
  - Subscripts follow the above rules as well
  - **Decimal point** (Hungarian students be careful!)
  - Known mathematical functions in upright font
- Auxiliary evaluations (if any) might be included as an appendix in the lab report. Note that the appendix is not necessarily corrected by the lab reader, it is only checked on demand.
- **Plagiarism meets zero tolerance (reference your sources).**
  - **No AI use for scientific content generation (if you don't learn why do you attend the uni?)**
  - **Zero score (first time)**
  - **Disqualification from the course (repeated case)**
  - **If data needed from others state that** (We forgot to measure the temperature, so we asked for the measured value from measurement pair X)

Structured ideas:

1. Title, header (Measurement pair code, Names, Date)
   a. Example:

      Thermal expansion - measurement report

      T5_1, Dávid Szaller & András Halbritter

   b. 07.02.2025

2. Abstract
    a. Attracts the reader's interest
    b. Free of technical details
    c. Briefly states the objective and results
    d. 2-5 sentences
        i. Example:
        In this experiment, we investigated the thermal expansion of the oil used
        in thermostatic valves. These valves convert changes in the oil's volume into the linear motion of a piston. To analyze this behavior, we
        measured the piston's position across a wide temperature range and
        calculated the oil's cubic thermal expansion coefficient
3. Introduction (to be neglected for physicist-engineers)
4. Elaboration of each task
    a. Structure for each task:
        i. 2-3 sentence description of the task "what are we doing?" (kind of an abstract for each measurement)
        ii. Brief theory (not copied from the lab description) essential for interpreting the results (reference the lab description)
        iii. Description of the experimental method and setup
        iv. Presentation of measured/calculated results (in tables, figures, text...). (figures are preferred)
        v. Brief discussion of results
b. Calculations
    i. Reference formulas used (lab description/elsewhere)
    ii. Number formulas
    iii. Error calculation where relevant (error propagation)
    iv. Error estimation
        1. Example:

        We used the caliper to measure the diameter of the piston, $d = 2 \pm 0.05$ mm, while the oil volume at room temperature was provided in the measurement handout, $V = 1 \pm 0.1$ cm^3. Using these values, we determined the cubic thermal expansion coefficient as $\beta = 7.2 \cdot 10^{-4} \frac{1}{°C}$.

        To estimate the uncertainty in β, we applied the variance method of error propagation. This approach is valid in our case since the errors of the different quantities in the formula for β are uncorrelated. As detailed in the Appendix, we obtained an approximate relative error of 11%. The dominant source of uncertainty in our measurement is the 10% error of the oil volume value.

2. Details are in the lab notes (if not, include in the report but well structured):

As the error of *b* is dominated by the error of *V*, at this error estimation we neglect the effect of the errors of *d* and that of the fit

```
b_error=b*V_error/V
```

```
b_error = 7.1982e-05
```

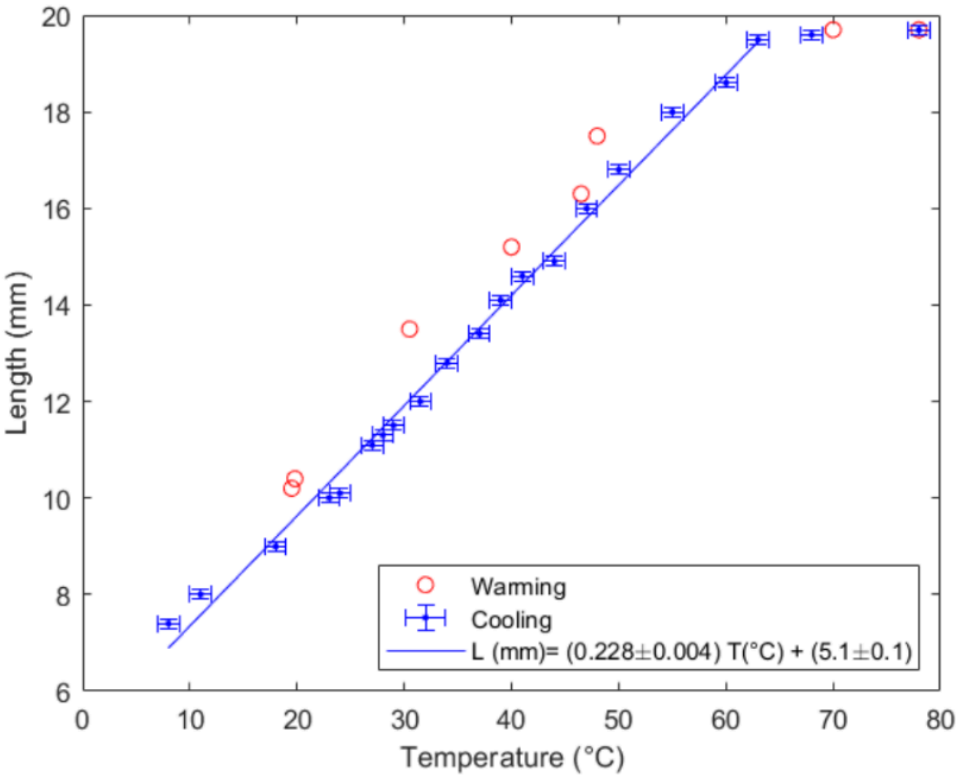
3. Or in the appendix:

Error calculation using the variance formula

```
%as d is quadratic in the formula of b, the relative error of d is multiplied by 2
b_error=b*sqrt((V_error/V)^2+2*(d_error/d)^2+(p1_error/p1)^2)
```

```
b_error = 7.6947e-05
```

c. Figures
   i. Figure labels are readable size, ideally matching text size
   ii. Legend if multiple symbols are present on the figure
   iii. Ticks, units, quantities present on axes
   iv. Few data points shown with markers (or connected points if showing a trend), continuous measurements shown with lines
   v. Errorbar if relevant
   vi. In case of fitting:
      1. Formula of the fitted curve
      2. Fitting result (can also go in figure caption)
      3. Fitting method, fitting error
   vii. Figure example:



viii. The **caption** briefly explains what is on the figure (usually placed under the figure). Legend can even be replaced by it.
Example: Fig.3. Temperature dependence of the piston length extending from the valve body. Red and blue symbols represent the measurements taken during the warming and cooling processes, respectively. For the cooling process, the uncertainty in the measured temperature and length values is indicated by horizontal and vertical error bars. The blue line represents the linear fit for the linear regime of the cooling dataset, accounting for measurement errors.

ix. Multi-panel figures: a), b), etc. panel labels instead of "on the top left figure"

d. Text
   i. Not bullet points, but continuous prose text
   ii. English style/grammar can be improved with AI, but do not generate results with AI

e. Brief discussion of results for each task
   i. Comparison with literature data
   ii. Comparison with expectations described in the lab notes
   iii. Expected and observed inaccuracies/errors, their causes
   iv. Example:

> ### Conclusion
>
> By utilizing a narrow piston to convert the thermal expansion of the oil in the thermostatic valve into an easily measurable linear motion, we determined the cubic thermal expansion coefficient of the oil. Our result, $\beta = (7.2 \pm 0.8) \cdot 10^{-4} \frac{1}{°C}$, is in good agreement with the literature value[1] for paraffin oil, $7.64 \cdot 10^{-4} \frac{1}{°C}$.

5. Summary (for the whole report if applicable)
   a. Comparison of own results with literature values (**lab description**/other public data)
   b. Discussion of other conclusions
6. Bibliography, references
   a. Literature used (e.g. lab description)
   b. Literature used for relevant comparisons with literature values
   c. Use an automatic citation manager; reference format should be uniform

Lab Notes

1. Makes it possible to write a report from the measurement (even weeks later)

2. Starting: Names, Neptun codes, course code, number of the measurement pair, date, title of the measurement
3. Text content:
    - Raw data if few (saved in tables/variables)
    - Type of instruments/sensors
    - Settings of instruments, readings of all available sensors
    - Wiring diagrams/experimental setup (take photos)
    - **Estimated errors (instruments/methods)**
4. Figure content
    - Raw data plotted (if many measurements were done)
    - Semi-processed data plotted (to catch the trends)
    - Dimensions/measured quantity
5. Initial evaluations
    - Rough estimates to catch errors
    - Do initial evaluations to catch errors
    - Write short codes for that